

THAARU Sepsis AI — Deep Learning Benchmarks

Phase 7 Deep Learning Report • Generated: July 28, 2026

1. Executive Summary

We constructed sequential neural network frameworks including LSTM, GRU, BiLSTM, and Transformers. Models were optimized via weighted BCE loss to handle class imbalance, and early stopping monitored validation AUPRC directly to prevent overfitting.

2. Deep Learning Leaderboard

Model	Config	AUROC	AUPRC	F1-Score	Recall	Parameters
LSTM	w12_h0	0.7450	0.0970	0.1446	0.3221	150,337
GRU	w12_h0	0.7764	0.0947	0.1422	0.4381	112,769
BiLSTM	w12_h0	0.7931	0.0951	0.1351	0.5391	333,441
Transformer	w12_h0	0.7888	0.0987	0.1029	0.7106	92,097
GRU (Selected)	w6_h0	0.7675	0.0828	0.1066	0.5665	112,769
GRU (Selected)	w24_h0	0.7922	0.1157	0.1551	0.5480	112,769
GRU (Selected)	w12_h3	0.7528	0.0852	0.1368	0.4881	112,769
GRU (Selected)	w12_h6	0.7722	0.1018	0.1299	0.5204	112,769
BiLSTM (Final)	w12_h0	0.7899	0.1238	0.1288	0.5669	333,441
BiLSTM (Final)	w6_h0	0.7898	0.0936	0.1144	0.6268	333,441
BiLSTM (Final)	w24_h0	0.7667	0.1040	0.1590	0.4899	333,441
BiLSTM (Final)	w12_h3	0.7666	0.0951	0.1130	0.5833	333,441
BiLSTM (Final)	w12_h6	0.7529	0.0795	0.1780	0.2847	333,441